Dashiell Kosaka

Email: dashkosaka@gmail.com
Mobile: +1 (312) 203-7512
Website: dashkosaka.github.io
LinkedIn: linkedin.com/in/dash-kosaka
Residence: Austin,Texas

Skills

- **ML/DL/CV**: **PyTorch**, Cupy, Keras, TensorFlow, Scikit-learn, Scipy, Numpy, Pandas, RAG, LLM, RNN, GAN, CNN, Reinforcement Learning, Deep Q Learning, Clustering, Segmentation, Detection, Bayesian Modeling, OpenCV, SIFT, RANSAC, OCR, Regression, Classification
- **Robotics**: ROS, Embedded Systems, Microcontrollers, Multi-Object Tracking, Kalman Filtering, Particle Filtering, SLAM, Data Association, LAP, Occupancy Grid Mapping, Radar, Camera, Sensor Fusion, Perception
- **Cloud & DevOps**: Agile, Bitbucket, Confluence, Jira, CI/CD, **Git**, Mercurial, AWS, **GCP**, Docker
- **Programming**: **Python**, C, **C++**, WSL, MATLAB, CUDA, x86, RISC-V, **Bash**, JavaScript, HTML, SystemVerilog
- **Other**: Data Analysis & Visualization, Scripting & Automation, **Prototyping**, Signal Processing, Remote Sensing
- **Languages**: English (Native), Japanese (Conversational), Spanish (Basic)

Work Experience

Senior Systems Engineer – Algorithm & Radar Feature Development
Uhnder Inc. – Austin, TX | January 2023 – Present

- **Prototyped Python and MATLAB solutions** for 5 radar interference management algorithms, **boosting interference resilience by over 10x** and restoring more than 20 dB of target SNR
- Integrated interference management algorithms into a **real-time C++ embedded radar system**, meeting strict 20 Hz operational requirements under memory constraints
- Architected an analytics framework to identify edge cases across 8 production automotive radar modules, **improving stability against millions of radar sensors** in use today
- Automated system-level testing for interference sensing and mitigation APIs, accelerating deployment cycles and **enhancing testing efficiency by 500%** through Python, Bash, Bitbucket pipelines, and CI/CD best practices
- Presented radar feature developments to potential investors, **securing multimillion-dollar investments** by showcasing algorithmic improvements

Systems Engineer – Algorithm & Radar Feature Development
Uhnder Inc. – Austin, TX | June 2020 – December 2022

- **Deployed and validated 4 radar perception algorithms** across teams, ensuring seamless integration into our ROS stack to enhance system performance and reliability in an **automotive environment**
- Implemented a **scalable multi-object radar tracking system**, delivering a C++ API for production use on tens of thousands of radar modules
- Modeled a Bayesian mapping algorithm for **sensing static and dynamic targets in autonomous systems**, further **enhancing runtime performance by 90% through CUDA** and architectural optimizations

Academic Research Intern – Remote Sensing & ML/CV
National Center for Supercomputing Applications – Champaign, IL | October 2019 – June 2020

- Engineered a **U-Net model in a deep learning pipeline** for detecting waterlogged farmland, **attaining 95% segmentation accuracy** on 3-meter resolution PlanetLab CubeSat imagery
- Expanded the waterlogging detection solution from Champaign County to hundreds of counties across Illinois, processing **large datasets with multi-channel GIS data**
- Resolved extrinsic calibration errors in satellite imagery, **achieving sub-meter accuracy** using SIFT, RANSAC, and geometric correction techniques

Software Engineering Contractor – Embedded Software & Front-End Development
Cohesive Manufacturing – Champaign, IL | September 2019 – March 2020

- Led the development of an embedded Arduino system for sensor data logging, creating drivers for 3 mission-critical sensors and **optimizing cloud logging with sub-20 ms latency for real-time monitoring**
- Enhanced customer website engagement by implementing multi-language support, boosting usability and increasing global reach for manufacturing partners

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Projects

Uhnder Interference Manager
Uhnder Inc. | Nov 2021 – Present
- Investigated and deployed **signal processing algorithms to detect, classify, and mitigate interference**
- Evaluated algorithms on an embedded radar system, establishing KPIs to measure performance improvements

Uhnder Object Layer
Uhnder Inc. | June 2020 – Oct 2021
- Explored perception-related radar algorithms to enhance output through refinement, tracking, and mapping
- **Facilitated radar feature integration by prototyping ROS nodes** and merging them into the ROS stack

OpenRadar [github.com/PreSenseRadar/OpenRadar | arxiv.org/abs/1912.12395]
PreSense | January 2019 – June 2020
- Oversaw the **development of an open-source radar signal processing library**, cited globally by researchers
- Contributed tools for generalized radar ADC data processing and AI-related functionalities

AlphaSmash [github.com/DashKosaka/AlphaSmash]
HackIllinois 2019 Hackathon | February 2019
- Developed a **Double Deep Q Learning (DDQN) agent** to autonomously play Super Smash Bros. Ultimate
- Assembled a hardware interface using an Arduino and camera to interpret state space and execute actions

Deep Fake Farms
AGCO Accelerator 2019 Hackathon | February 2019
- Utilized a **GAN in PyTorch to synthesize agricultural imagery**, enabling large-scale data augmentation
- Created a scalable generation method capable of producing arbitrarily sized outputs

EIE.io
AGCO Accelerator 2019 Hackathon | February 2019
- Built an **object detection pipeline with TensorFlow and React** to monitor livestock health
- Leveraged GCP's Vision API and Firebase Realtime Database to store data and alert users of unhealthy animals

TyperML
UIUC Personal | December 2018
- Designed an interactive typing game to improve users' typing skills by learning commonly missed sequences
- **Implemented RNN models to generate text sequences while predicting typing speed and accuracy**

RISC-V Architecture [github.com/DashKosaka/RISC-V]
UIUC ECE | Top 5 | October 2018 – December 2018
- Pipelined a 5-stage RISC-V CPU, optimizing throughput and latency while executing on an FPGA
- Enhanced the architecture with a PHT branch predictor and L3 cache, earning a top 5 competition finish

SrirachOS Operating System [github.com/DashKosaka/SrirachOS]
Microsoft Operating System Design Competition | Top 4 | February 2018 – May 2018
- Engineered the file system architecture, pagination, and context switching for a CLI-based OS
- Upgraded the system with a search function and command history, improving user experience and efficiency

FPGA Stepmania [github.com/DashKosaka/fpga_stepmania]
UIUC ECE | December 2017
- Orchestrated a state machine in SystemVerilog to manage gameplay logic for a StepMania variant
- Devised a VGA display driver to render game assets directly on-screen using the Altera DE2-115 FPGA

Education

B.S. Computer Engineering with Honors
University of Illinois at Urbana-Champaign | September 2015 – May 2019
- **Artificial Intelligence**, **Deep/Machine Learning**, **Computer Vision**, **Data Structures & Algorithms**
- Operating Systems, Probability & Statistics, Linear Algebra, Signal Processing, Computer Architecture